

Lecture 1: Foundations

Probability vs Statistics · Population & Sample · i.i.d. · Plug-in Principle · Loss & Risk

How much should you trust a number?

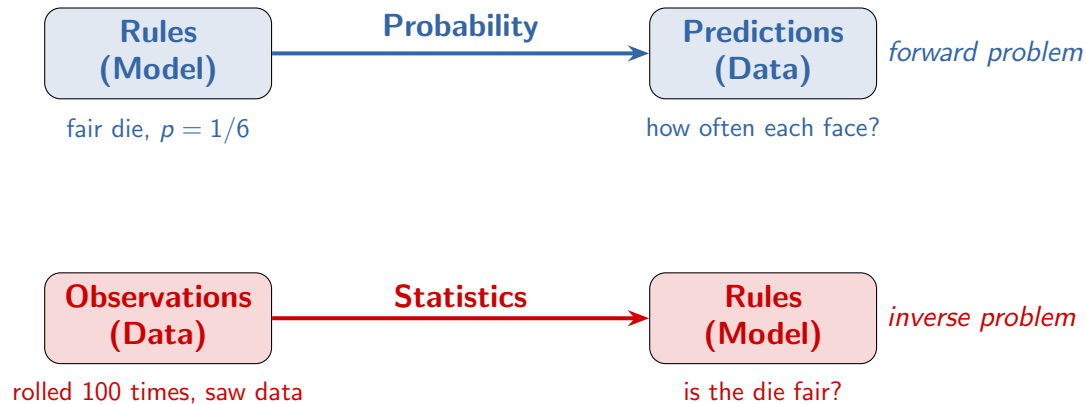
A poll says: “52% support candidate A” ($n = 1,000$)

A clinical trial says: “Drug B reduces symptoms by 15%” ($n = 200$)

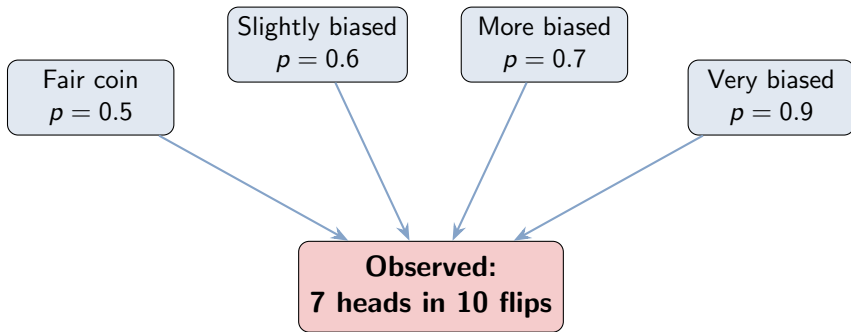
How confident should we be?

This entire course is about answering this question rigorously.

Probability vs Statistics



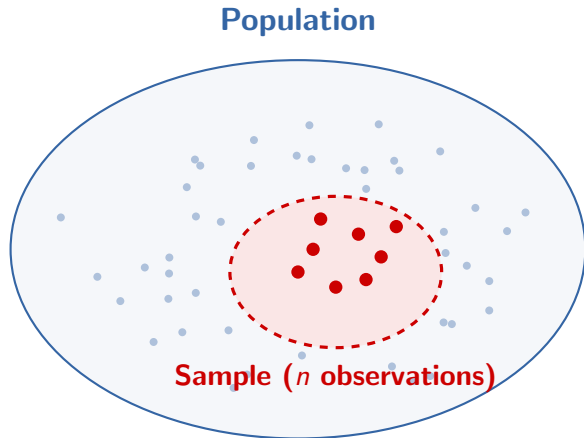
Why the inverse problem is harder



Many different models could have produced this data!

The inverse problem is **ill-posed** — statistics gives us tools to navigate this.

Population vs Sample



Population:

All units of interest

Can be finite or
conceptually infinite

Sample:

The subset we
actually observe

Parameter vs Statistic

Parameter θ

Fixed, unknown number
Describes the **population**

Examples:

μ = true mean lifetime

p = true approval rate

σ^2 = true variance

we estimate this
using this

Statistic $T(X_1, \dots, X_n)$

Random variable, computable
Computed from the **sample**

Examples:

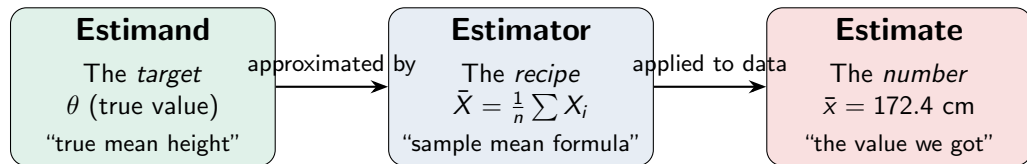
$\bar{X}$ = sample mean

$\hat{p}$ = sample proportion

S^2 = sample variance

A **parameter** is a fixed number. A **statistic** is a random variable.
Confusing these is the source of most beginner mistakes.

The Triple: Estimand / Estimator / Estimate



A polling agency surveys 1,000 people and reports:

“62% support policy X”

Identify each:

1. What is the **population**?
2. What is the **parameter**?
3. What is the **sample**?
4. What is the **statistic**?
5. What is the **estimate**?

Discussion: Answers

“62% support policy X” ($n = 1,000$)

1. **Population:** all citizens of the country (eligible voters)
2. **Parameter:** p = true proportion who support policy X (unknown)
3. **Sample:** the 1,000 people surveyed
4. **Statistic (estimator):** $\hat{p} = \frac{\# \text{ who said "yes"}}{n}$ (the formula/recipe)
5. **Estimate:** $\hat{p} = 0.62$ (the specific number from this sample)

The i.i.d. Assumption

Classical statistics assumes our sample $X_1, X_2, \dots, X_n$ is **i.i.d.**:

Independent

Knowing X_1 tells you
nothing about X_2

Each observation is a fresh draw

Identically Distributed

Every X_i comes from the
same distribution F

Same process generates each one

When does i.i.d. hold?

- ✓ Random sampling from a large population
- ✓ Repeated independent measurements of the same quantity
- ✓ Controlled experiments with proper randomization

i.i.d. is an **idealization** — it's approximately true in many practical settings, and most of what we'll do this course assumes it.

When does i.i.d. break?

Time dependence

stock prices, weather

Non-response bias

who refuses the survey?

Spatial correlation

neighboring sensors

Distribution shift

training data $\neq$ deployment

Selection bias

hospital-only patients

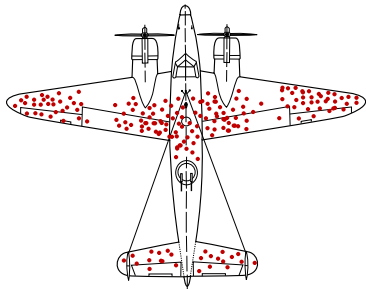
Clustering

students within schools

Not a disaster — just means you need different tools.

But if you *pretend* non-i.i.d. data is i.i.d.,
your conclusions can be **wildly wrong**.

Survivorship Bias



WW2: Engineers studied bullet holes on returning bombers and proposed armoring the hit areas.

Abraham Wald: *“You’re only seeing planes that **survived**. Armor the places with **no** holes — those hits brought planes down.”*

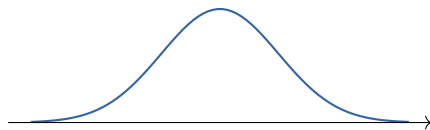
More examples:

- ▶ Online survey: “Do you have internet?” — 100% say yes
- ▶ “Soviet products lasted forever” — you only see the ones that survived
- ▶ Bus fare survey: asking people *on the bus* “100→150 AMD?” — only sampling current riders

The Plug-in Principle

Idea: We don't know the true distribution F , so replace it with the **empirical distribution** $\hat{F}_n$.

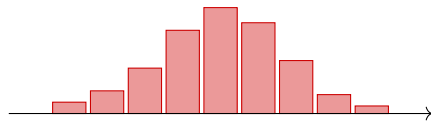
True distribution F
(unknown)



smooth, continuous

replace with
→

Empirical distribution $\hat{F}_n$
(computable from data)



mass $1/n$ on each point

Plug-in in Action

Replace the **population quantity** with its **sample analogue**:

Want	Population	Plug-in
Mean	$\mu = \mathbb{E}_F[X]$	$\hat{\mu} = \bar{X} = \frac{1}{n} \sum X_i$
Variance	$\sigma^2 = \text{Var}_F(X)$	$\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$
CDF	$F(t) = P(X \leq t)$	$\hat{F}_n(t) = \frac{\#\{X_i \leq t\}}{n}$

Glivenko–Cantelli theorem: $\hat{F}_n \rightarrow F$ uniformly as $n \rightarrow \infty$.

(The “fundamental theorem of statistics” — connects to LLN from Module 20.)

The Summarization Problem

You must summarize a distribution with a **single number** a .
How do you choose?

It depends on what “error” means to you.
This is formalized by a **loss function** $L(\theta, a)$.

Three Losses, Three Optimal Summaries

Squared Error

$$L = (\theta - a)^2$$

Penalizes large errors heavily



Mean

Absolute Error

$$L = |\theta - a|$$

Linear penalty, robust to outliers



Median

0-1 Loss

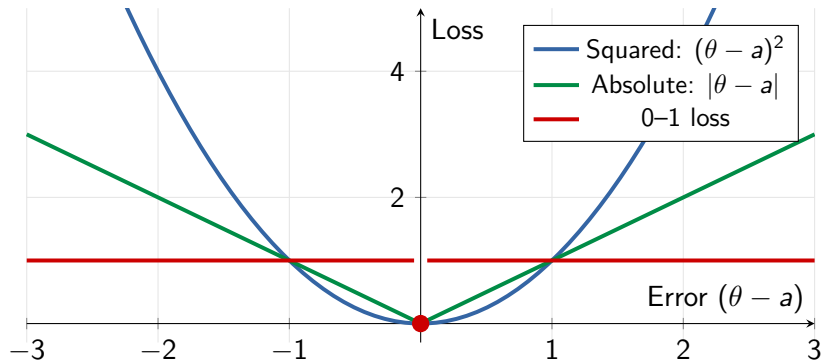
$$L = \mathbf{1}[\theta \neq a]$$

Wrong or right, nothing in between



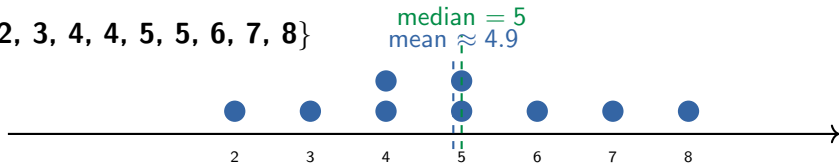
Mode

Visualizing the Losses

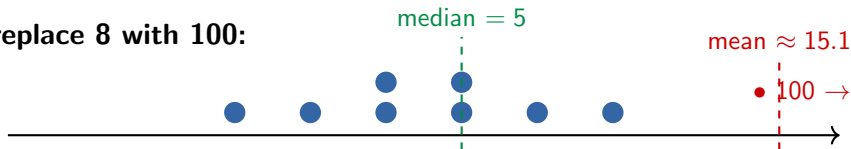


Mean vs Median: Sensitivity to Outliers

Dataset: {2, 3, 4, 4, 5, 5, 6, 7, 8}



Now replace 8 with 100:



One outlier moved the mean from 4.9 to 15.1.
The median didn't budge.

The Mean Can Mislead

Three statisticians go hunting.

They spot a deer. The first one fires and misses **5 meters to the right**.

The second one fires and misses **5 meters to the left**.

The third one exclaims: *"We got him!"*

Average diet.

If one class of people eats **tup** and another eats **meat**,
then on average everyone eats **tolma**.

Risk and Empirical Risk

Risk (theoretical)

$$R(\theta, \hat{\theta}) = \mathbb{E}[L(\theta, \hat{\theta})]$$

Average loss over
all possible samples

(unknown — depends on F)

approximate
----->

Empirical Risk

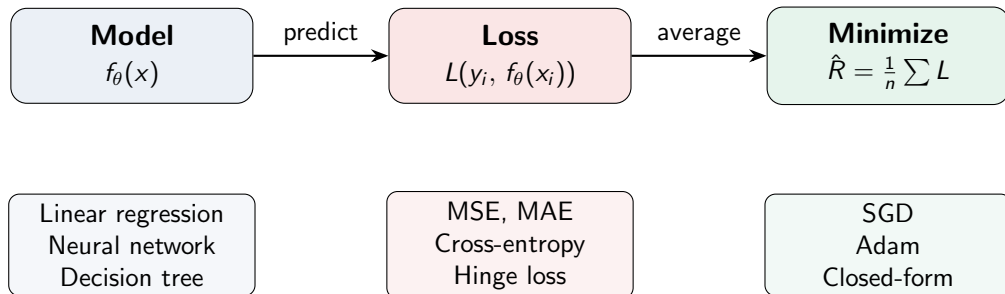
$$\hat{R} = \frac{1}{n} \sum_{i=1}^n L(X_i, a)$$

Average loss on the
data we actually have

(computable!)

Empirical Risk Minimization (ERM): choose the estimator that minimizes $\hat{R}$.
This principle unifies least squares, maximum likelihood, and most learning algorithms.

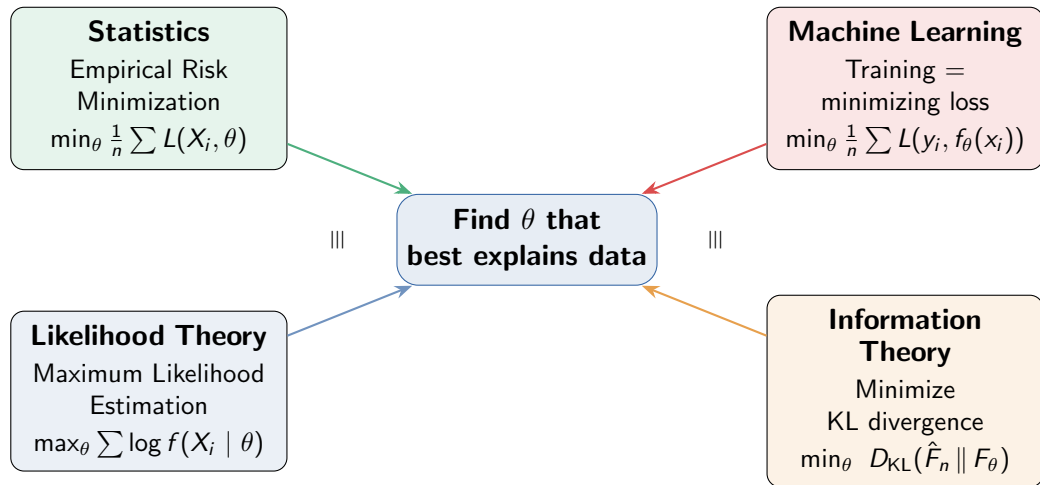
ERM in Machine Learning



Every ML training procedure is ERM.

Choose a model class, choose a loss, minimize the empirical risk over parameters.

One Principle, Many Names



MLE with log-loss = **ERM** with neg. log-likelihood = **minimizing** KL divergence to data.

Homework

1. A factory claims its light bulbs last 1000 hours on average ($\mu = 1000$). You test 50 bulbs and find $\bar{X} = 980$.
Identify: (a) the population, (b) the sample, (c) the parameter, (d) the statistic, (e) the estimator, (f) the estimate.
2. For data $X = \{1, 2, 3, 4, 100\}$:
 - (a) Find the value c that minimizes the empirical risk under L_2 loss.
 - (b) Find the value that minimizes L_1 loss.
 - (c) Explain which loss function is more appropriate here and why.
3. For data $X = \{2, 3, 5, 7, 8\}$, use the plug-in principle to estimate:
 - (a) the mean, (b) the variance (using $\hat{F}_n$, dividing by n), (c) the median, (d) the 75th percentile.

Questions?

Next lecture: Descriptive Statistics & Empirical Distributions